ITA0612 – MACHINE LEARNING

ASSIGNMENT – SGD GOALS

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- Design a fish species classification system using Deep Learning.

1. Problem Statement

Accurate identification of fish species is essential for:

- Marine biodiversity conservation
- Sustainable fisheries management
- Monitoring endangered species
- Preventing illegal fishing activities

2. Objective

To develop a Deep Learning model that automatically identifies fish species from underwater or captured fish images and supports marine ecosystem conservation under **SDG 14 – Life Below Water**.

3. System Architecture

```
Fish Image Dataset
          |
          ▼
 Image Preprocessing
(Resize, Normalize)
          |
          ▼
 Data Augmentation
(Rotation, Flip, Zoom)
          |
          ▼
 Deep Learning Model
       (CNN)
          |
          ▼
 Feature Extraction
          |
          ▼
 Species Classification
          |
          ▼
 Predicted Fish Species
```

4. Dataset

Possible datasets:

- Fish4Knowledge Dataset , Kaggle Fish Species Dataset , Large Scale Fish Dataset

Dataset contains:

- Fish images , Species labels , Different orientations and backgrounds

Example classes:

- Tuna , Salmon , Tilapia , Catfish , Mackerel , Sardine

5. Data Preprocessing

Step 1: Image Resizing

Resize all images to: 224 × 224 pixels

Step 2: Normalization

Pixel values: 0 – 255 → 0 – 1

Step 3: Data Augmentation

Techniques:

- Rotation , Horizontal Flip , Zoom , Brightness Adjustment

Purpose:

- Increase dataset size , Reduce overfitting

6. Deep Learning Model

Convolutional Neural Network (CNN)

Layer 1

- Convolution Layer , ReLU Activation , Max Pooling

Layer 2

- Convolution Layer , ReLU Activation , Max Pooling

Layer 3

- Fully Connected Layer

Output Layer

- Softmax Classifier

7. Training Process

Input

Fish images

Training

Forward Propagation - Loss Calculation – Backpropagation - Weight Update (Adam Optimizer) - Repeat for Multiple Epochs

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8. Performance Parameters

Parameter	Purpose
Accuracy	Overall classification performance
Precision	Correct positive predictions
Recall	Ability to identify species correctly
F1-Score	Balance of precision and recall
Confusion Matrix	Detailed classification results
Loss	Training performance

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9. Expected Results

Metric	Expected Value
Accuracy	90–98%
Precision	>90%
Recall	>90%
F1-Score	>90%

10. Applications

- Marine biodiversity monitoring
- Smart fisheries management
- Ocean ecosystem assessment
- Endangered species protection
- Underwater robotic exploration
- Fish market automation

11. SDG 14 Connection – Life Below Water

This project contributes to **Sustainable Development Goal 14** by:

- Conserving marine biodiversity
- Monitoring fish populations
- Supporting sustainable fishing practices
- Protecting endangered aquatic species
- Improving ocean resource management